

Enhancing Adaptive Personalized Learning Interfaces with Generative AI for Individuals with ADHD

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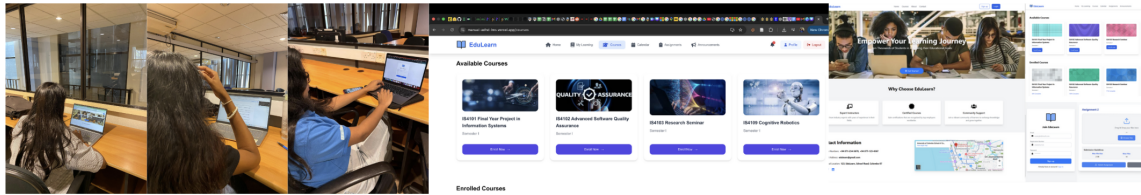


Figure 1: Prototype testing of Learning Management Systems through User-Centered Design

Abstract

Students with Attention deficit hyperactivity disorder (ADHD) face unique challenges in educational environments due to their impulsivity, hyperactivity, and distractibility. Conventional learning systems often do not cater to the specific needs of ADHD learners, indicating a demand for tailored ADHD-friendly educational platforms that provide these students with equitable opportunities to excel academically. To address this research gap, we explored how Generative AI could enhance personalized learning interfaces for ADHD students aged 18–30 who have experience with online learning. This initiative aims to create an innovative learning platform for ADHD individuals to enhance their motivation and academic success. We employ statistical analysis methods to evaluate three different learning management systems (LMS): a general-purpose LMS, a manually optimized LMS for ADHD users, and an AI-driven LMS with improved prompting features. We evaluated the feedback from 22 ADHD students to assess measures of the usability and effectiveness of each system through the System Usability Scale (SUS), the User Experience Questionnaire (UEQ), task analysis, and evaluations by UI/UX professionals and HCI experts to gauge learning outcomes and engagement of users. The research discovered that users prefer the AI-generated LMS for usability and the manual system for effectiveness. The findings indicate that although AI-generated user interfaces (UIs) enhance usability, human-designed

UIs are essential for educational effectiveness. The study promotes hybrid methodologies that combine AI efficiency with educator-led instructional design, providing practical insights for user interface (UI) developers and institutions focused on achieving a balanced user experience (UX) and learning outcomes.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; Accessibility design and evaluation methods; • **Applied computing** → Computer-assisted instruction.

Keywords

User Interfaces, Accessibility, User experience, Artificial Intelligence, Attention deficit hyperactivity disorder (ADHD)

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1 Introduction

According to Furman [15], ADHD is a common neurodevelopmental disorder, which affects children's ability to control their spontaneous reactions, including movement, speech, and attentiveness. ADHD implies a distinct neuro-divergent method of the human brain's function which differs from neurotypical people. It usually appears in childhood and persists into adulthood and can be identified with the persistent and pervasive symptoms of inattention,



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hyperactivity, and impulsivity [45]. Erskine et al. [13] state that long-term ADHD is associated with a high risk of mental disorders, interpersonal issues, and academic setbacks.

A significant number of children with ADHD tend to have particular learning disabilities, which may lead to difficulties in reading, writing, and mathematics [3]. According to the growing prevalence of ADHD diagnoses and the unique challenges encountered by people with them, there is an urgent need to improve specific educational tools that enable personalized learning for them. Different studies have been conducted to improve personalized learning for neurodivergent students, there is a lack of studies in the area of enhancing personalized learning interfaces specific for ADHD individuals. Although similar studies like Gkora [16] have been conducted to support the primary education of ADHD students, there is still a significant lack of research in the area of addressing the specific learning requirements of adult ADHD individuals.

Our research employs Generative AI (GenAI) to create personalised user interfaces for individuals with ADHD and identify the needs and preferences of ADHD learners and develop UI designs that improve engagement, focus, and learning effectiveness. This research seeks to address the needs of ADHD students by designing UI based on user feedback, usability heuristics, and Generative AI with the aim of create an accessible and engaging learning environment for ADHD students, enhancing academic outcomes and learning experiences.

1.1 Background

1.1.1 What is ADHD? According to Diagnostic and Statistical Manual of Mental Disorders V (DSM-V), mental disorders that manifest during childhood or adolescence can be divided into three main categories which consist of Neurodevelopmental disorders (NDDs), Autism spectrum disorders (ASDs) and Mental health disorders (MHDs). ADHD is a common Neurodevelopmental disorder that significantly affects people's ability to perform their daily tasks. NDDs have been noted as a major reason for childhood disabilities, impacting not only the affected child but also their loved ones and society as a whole. As stated in Barua et al. [5], it is estimated that between 30% and 50% of kids with ADHD have co-occurring mental health disorders, like depression. It indicates that NDDs and MHDs often overlap with one another. Such complex combinations of comorbidities between MHDs and NDDs may lead to learning disabilities and negatively impact a person's quality of life [5]. Because of that, if an individual is suffering from ADHD along with MHDs, it affects them worse. Therefore, an advanced approach is required to enhance personalized learning for individuals with ADHD, considering their cognitive and attentional differences.

ADHD is a complex disorder influenced by both genetics and environmental factors. The disorder's exact cause is unknown, but studies suggest it is mainly influenced by genetics and the physical environment. Non-genetic factors such as brain injuries, early birth, maternal smoking, and exposure to environmental toxins like lead are also linked to ADHD. According to the study, Ibrahim et al. [20] which focuses on the ADHD disorders of students between 6-18 years old, there are three forms of ADHD that are prevalent in kids and students.

Table 1: ASRS v1.1 scale scoring

ADHD Score	ADHD Range
0–9	Low negative
10–13	High negative
14–17	Low positive range
18–24	High positive range

- Inattentive type: Student is unable to focus or maintain concentration on an assignment or activity. It appears more common in girls to have this disorder.
- Hyperactive-impulsive type: This type of student exhibits excessive behavior and occasionally acts without thinking, but they do not display severe issues with attention. It appears more prevalent among boys to have this disorder.
- Combined type: The learner exhibits impulsivity, hyperactivity, and inattention.

1.1.2 ADHD Severity scales and avoiding associated risks. Based on symptom severity and comorbidity, ADHD can range from mild to severe. Assessing severity helps reduce high-risk population risks and customise ADHD education. When conducting research based on feedback and user evaluations from ADHD students to enhance their learning, high-risk individuals who may suffer from complex comorbidities may not be able to participate in user evaluations due to associated risks.

Therefore we can use the Adult ADHD Self-Report Scale (ASRS v1.1) [34], which is a commonly used tool to assess ADHD severity of adults, developed in collaboration with the World Health Organization (WHO). It is an 18-item self-report questionnaire designed to assess Attention Deficit Hyperactivity Disorder (ADHD) symptoms in adults (18+). Based on the WHO Composite International Diagnostic Interview (2001), this scale's questions are tailored to address adult symptom manifestation and are in line with both the DSM-IV and DSM-5-TR criteria [21]. Using the symptom checklist of ASRS v1.1 we can evaluate the level of impairment of the ADHD individuals associated with the symptoms and consider the symptom severity according to symptom frequency. By using 0-24 scale scoring method participants can be selected as in table 1.

Based on the studies like Kessler et al. [21], this Low Negative (0-9) range ADHD individuals exhibit minimal symptoms with negligible impact on daily functioning. High Negative (10-13) range individuals may suffer with noticeable symptoms, but they are still manageable without severe interference in daily activities. In the low positive (14-17) range, symptoms are more prominent, potentially affecting focus, impulsivity, and hyperactivity, requiring targeted support. Individuals belonging to the high positive (18-24) range are more likely to suffer from severe symptoms often associated with co-occurring other mental health conditions like depression, anxiety, insomnia, panic disorder and face severe difficulties with managing even daily activities.

Therefore, high positive range individuals above 18-24 are not encouraged to participate in the evaluations of our study. We selected the normal range (low negative, high negative, low positive range) ADHD individuals (0-17) to participate in the research and evaluated the user interfaces to determine what features will have the most impact for their educational activities.

1.1.3 Educational difficulties caused by ADHD. Students with ADHD often have short attention spans and are easily distracted [20]. ADHD symptoms include inattention, anxiety, and impulse control significantly impact academic performance and the school environment, making ADHD youngsters struggle in school[20]. Additionally, they struggle with task organisation and time management[20].

Ferguson et al. [14] Research indicates that university students with ADHD typically remain uninformed, humiliated, and hesitant to seek help, resulting in poorer GPAs compared to those without ADHD. ADHD undergraduates must deal with disorder symptoms and a new social milieu with less parental control compared to their childhood. Refusing help causes stress and isolation. Impulse control disorders make focus and job completion challenging. ADHD kids experience greater problems including new separateness, stress, complex study schedules, and less supports, which lowers self-esteem due to high symptoms and executive functioning. ADHD students have worse academic performance and greater dropout rates due to these educational problems.

1.1.4 Importance of digital learning tools. Digital learning tools are vital for supporting students with ADHD by offering structured, engaging content and aids for organization and time management, which improve focus and academic performance. AI technologies like large language models and generative AI further enhance these tools by enabling personalized, adaptive learning experiences. Studies show that AI tools such as ChatGPT can effectively assist ADHD therapy and generate customized educational materials tailored to individual needs [7, 38]. Thus, AI-powered digital tools significantly improve learning outcomes for ADHD students.

1.2 Problem Statement

Students with Attention Deficit Hyperactivity Disorder (ADHD) encounter unique challenges in educational settings, including difficulties with impulsivity, hyperactivity, and inattention. Even though there are many conventional online learning platforms available, the majority are unable to meet the unique behavioral and cognitive requirements of ADHD students. A clear research gap can be identified which is still unexplored that is addressing the special learning abilities of ADHD individuals through exploring personalized learning platforms for them. Even though some studies, like Gkora [16], have been done to help ADHD students in primary school, there is still a dearth of research on how to meet the unique learning needs of adults with ADHD.

With technological advancements, although most of the current research emphasizes gamification, AR, VR, and sensory tools, they do not directly address ADHD learners' needs within personalized learning environments. Generative AI shows immense potential, yet its application for creating ADHD-friendly LMS interfaces remains largely unexplored. When applying Generative AI in personalized learning, most of the studies like Pesovski et al. [38] which explores a tool for generating personalized learning materials using LLMs, focus on customization of learning content but not on generating personalized learning interfaces. These research gaps emphasize the significant need for optimized learning interfaces suitable for adult ADHD students to excel in their academics.

1.3 Research Questions

The purpose of this research paper is to conduct a comprehensive exploration of the areas addressed by following research questions:

1.3.1 Primary Research Question.

- **How can AI-driven customization improve the effectiveness and usability of LMS for individuals with ADHD compared to conventional and manually optimized LMS platforms?**

Our research focuses on creating three prototypes for ADHD students: a conventional LMS for the general population, a manually optimised LMS for ADHD users, and an AI-generated LMS customised to ADHD needs. By conducting a comparative analysis, we explore how AI-driven customization improves the effectiveness and usability of LMS comparing to the other two experiments of conventional and manually optimized LMS. The intention is to explore the best learning experience for the ADHD individuals to enhance their educational outcomes, utilizing the potential of generative AI.

1.3.2 Secondary Research Questions.

- How can user data be effectively used to evaluate the usability of conventional LMS to identify limitations and areas for improvement?
- What improvements are necessary in personalized learning systems to support ADHD students?
- How can prompting strategies in generative AI be optimized to create user interfaces that enhance engagement and usability for ADHD learners?
- How can the impact of GenAI-driven personalized learning interfaces on academic performance and usability for individuals with ADHD be assessed?

2 Literature Review

2.1 Personalized Learning for Neurotypical Students

Personalised learning allows students to reach their educational requirements and capacities by providing adaptable, personalised content based on their potential and skills. In an e-learning platform it is crucial to have personalized interfaces for users ensuring individualization and differentiation because the student group we are focusing on is heterogeneous[44]. When applying personalization it is important to identify the learner first based on learner's intelligence according to Gardner's "Multiple intelligences" theory and the proficiency level in order to Dreyfus's Proficiency stages to address each learner's unique learning desires and requirements [44]. Shih et al. [44] explores how an e-learning system has a personalized interface, enabling different student views/learning objects and activities by applying Gardner's "Multiple intelligences" theory, Dreyfus's Proficiency stages focusing on a heterogeneous student group.

Adaptive user interfaces can also improve student engagement and educational outcomes, boosting personalisation. Research has identified many methods for adapting user interfaces, such as AI,

UM, and HCI for multimodal interfaces [4]. Intersection of personalization and adaptivity improves the usability and effectiveness of user centered learning. Recently Learning management systems (LMS) play a crucial role in assisting personalized education. Among the many technological advancements, Adaptive User Interfaces (AUI) has become a key part in keeping students interested in their studies. Bagustari and Santoso [4] is a study which analyzes the development of Adaptive User Interfaces(AUI) in education 4.0 for improving Learning management systems (LMS) to facilitate the dynamic changes in education following the approaches of AI, UM, HCI and using multimodal interfaces applied to create AUI. Bagustari and Santoso [4] discusses the application of technological and Heutagogical (individual) aspects in their evaluations.

Many different studies have been conducted in different areas focusing on leveraging personalized education using various technological advancements. Alves et al. [1] discusses how "personality" is important as a differentiating factor in user interface designing. According to Alves et al. [1], personality is defined as the collection of habitual behaviors, cognitive and emotional patterns that result from biological and environmental influences and it is important for fulfilling diverse user needs in customizing user interfaces. It also discusses the application of user personality, focusing on improving Graphical User Interface(GUI) design exploring the application of hybrid mental models and introvert, extravert, neutral GUI.

Effective user interface design relies on best practices such as Nielsen's usability heuristics, user-centered design principles, and color theory [19, 30]. While AI-driven personalized learning platforms have rapidly evolved, adapting content based on user data, most are tailored for neurotypical learners. There remains a significant gap in platforms designed to meet the specific cognitive and attentional needs of individuals with neurodevelopmental disorders like ADHD.

2.2 Personalized Learning for ADHD students

ADHD symptoms including impulsivity, hyperactivity, and inattention must be considered while designing personalised learning programs. These provide unique obstacles in traditional educational settings. Personalised learning helps ADHD children with attention span and impulse control. Different research have examined how controlled, distraction-free, engaging, and manageable learning environments improve ADHD students' learning experiences.

Gkora [16] highlights the importance of integrating autonomy-supportive practices and technological interventions in primary education for students with ADHD, emphasizing the need for interdisciplinary approaches. A personalized AI-based system which analyzes contextual data, identifies user behavior patterns, and effectively visualizes the information to provide insights to ADHD students, has been explored through the study Ravishankar [43]. As stated in Ravishankar [43], the project has used an iterative Double Diamond framework for incorporating co-design techniques to explore the real life problems of ADHD and ensure that the concept is tailored to the user's needs.

Numerous studies have been conducted on how general users comprehend data visualizations. Tran et al. [49] emphasizes how individuals with ADHD interpret data visualizations and how their

comprehension may differ from the general users, by conducting a crowd-sourced survey involving 70 participants with and without ADHD. Through that they have found that participants' accuracy and response times are impacted by specific chart elements and ADHD. This study reveals that different chart embellishment types impact accuracy and response times for individuals with ADHD differently depending on the types of questions, suggesting visual design recommendations make accessible data visualizations for people with ADHD.

2.3 Existing learning applications for ADHD students

2.3.1 AHA: ADHD Augmented (Learning Environment). The AHA project, supported by the European Commission, intends to improve education for ADHD children by utilizing Augmented Reality and web technologies [19]. The project's goal is to improve reading and spelling skills while also increasing academic engagement by using augmented reality into online literacy programs. It aims to help ADHD students stay focused, make fewer mistakes, and complete homework more efficiently. It builds on established methodologies and works with the Web Health Application for ADHD Monitoring (WHAAM) project to ensure user privacy and data security[28]. AHA aims to give ADHD children a creative learning environment.

2.3.2 aTenDerAH: a videogame application to support e-Learning students with ADHD. aTenDerAH is a videogame that supports e-learning for young adults, particularly those with ADHD[26]. aTenDerAH was created with Unity, Cinema 4D for 3D modeling and animation, and Photoshop for texture creation. The video game was integrated into the Atutor e-learning platform to conduct a case study on the perception of aTenDerAH by students with ADHD, non-ADHD students, and teachers. Participants have expressed satisfaction with the tool's goals and the videogame's good impact on their learning process

2.3.3 The Facilitating On-going Concentration in Undergraduate Students (FOCUS) Device. As described by Thomas and Cezeaux [48], the FOCUS application is a prototype aimed at improving academic performance in students with ADHD by providing non-stigmatizing reminders to help maintain focus during class and study sessions. It utilizes the Texas Instruments eZ430 Chronos digital wristwatch to control desk peripherals like vibration motors, LEDs, and speakers, delivering discreet alerts through vibrations, lights, and sounds. Designed based on ADHD student feedback, the device is compact, lightweight, and unobtrusive, with randomized reminders to reduce habituation.

2.3.4 X5Learn: A Personalised Learning Companion at the Intersection of AI and HCI. X5Learn is a human-centered educational platform that uses AI and HCI to help users browse and access online educational resources [37]. Available at <https://x5learn.org>, it supports teachers in reusing and sharing course materials and offers students tools for content selection, note-taking, and reviews. Its personalized recommendation system adapts learning paths to user preferences.

2.3.5 Therapy Tool for Adolescents with ADHD. The Nugawela et al. [35] study developed an Android app to support ADHD treatment by tracking patient progress and symptoms. The app allows patients, parents, and doctors to monitor improvements, ensures accuracy by using individual testing modules, and addresses all ADHD symptoms. By tracking user performance and development, the app aims to improve treatment outcomes and help prevent ADHD symptoms from persisting into adulthood.

2.3.6 ECHO ADHD in India: a feasible and acceptable training model for child-care physicians to identify and manage attention deficit hyperactivity disorder. The ECHO ADHD model, piloted in India, is a tele-mentoring program designed to improve child-care physicians' ability to identify and manage ADHD. Through 14 interactive on-line sessions over six months, the program trained pediatricians and specialists using lectures and case discussions. The study found the ECHO ADHD model to be both feasible and acceptable, with significant improvements in participants' knowledge of ADHD diagnosis and self-efficacy. However, gains in knowledge about ADHD comorbidities and management were less pronounced. Overall, ECHO ADHD effectively builds physician capacity for ADHD care in low- and middle-income settings like India.

2.4 Use of GenAI in Personalized Learning

AI significantly enhances personalization in education, with Machine Learning (ML), Generative AI (GenAI), and Large Language Models (LLMs) increasingly powering personalized learning platforms. GenAI, in particular, enables dynamically adaptive educational experiences. Key model architectures include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion models. Tools such as GPT, Jasper, AI-Writer (text generation), and DALL-E 2, Midjourney, Stable Diffusion (image generation) are transforming educational content creation.

Several studies demonstrate GenAI's role in personalized education. Yu et al. [52] highlights how AI improves MOOCs by enabling personalized learning paths and non-linear content access. Perez-Ortiz et al. [37] explores a human-centered AI platform that enhances access to free educational resources. A systematic review by Castro et al. [10] outlines AI's role in reinforcing personalized learning through adaptive pedagogical strategies.

Pesovski et al. [39] presents a tool integrated into a college LMS that uses LLMs (via OpenAI's API) to generate class-specific materials based on instructor-defined outcomes. Logacheva et al. [24] reports positive results from a user study in a programming course using GPT-4 to generate context-aware exercises, noting increased student engagement. Berrezueta-Guzman et al. [8] investigates combining LLMs and robotics to support ADHD therapies in intelligent environments, discussing theoretical, practical, and ethical aspects.

Barua et al. [5] evaluates AI tools for students with neurodevelopmental disorders (NDD), showing benefits in social interaction and personalized education. Techniques include both traditional ML and advanced models like CNNs and LSTMs. Sukiman and Aziz [46] reviews AI-driven personalized interventions for students with learning difficulties, emphasizing implementation needs and essential features for success.

2.5 Use of GenAI in Interface Generation

Recent advancements in Generative AI have significantly enhanced UI generation capabilities. Goloujeh et al. [17] investigates user experiences with text-to-image tools, highlighting prompt creation and refinement through interviews with 19 users. Similarly, Rajcic et al. [42] explores how seven visual artists use prompt-based interfaces for ideation and production, emphasizing ethical design considerations for future GenAI tools. Experimental work by Liu and Chilton [23] identifies key hyperparameters and failure modes in prompt engineering across five studies, showing notable variation in output quality. Meanwhile, Weisz et al. [50] outlines six principles for GenAI application design, grounded in Nielsen's "intent-based outcome specification" framework [31]. These insights and principles inform our approach to generating adaptive user interfaces tailored for enhancing learning experiences in ADHD students.

3 Methodology

The research was conducted in 6 main phases, aligned with the Design Science Research Methodology (DSRM) [36], which is suitable for solving real-world challenges through development and evaluation of innovative artefacts [9]. The study aimed to create and evaluate personalised learning interfaces for ADHD students using generative AI. DSRM was chosen because it can provide both scientific rigour and practical relevance in Information Systems (IS) research [6]. The methodology made it easier to create and compare three different learning management systems: a traditional LMS, a manually optimised ADHD-specific LMS, and an AI-driven LMS with enhanced prompting.

3.1 Problem Identification and Motivation

The study identified a gap in personalized learning platforms for adults with ADHD, as most existing research focuses on early education. Despite the potential of generative AI, its application in ADHD-friendly LMS design remains unexplored. Surveys and interviews with medically and self-diagnosed adults (ages 18–30) confirmed the need for an AI-powered, personalized solution to support their academic success.

3.2 Defining Solution Objectives

Recognizing the limitations of traditional LMS platforms for ADHD students, the study aimed to develop a personalized, adaptive learning environment using generative AI. It focused on identifying the most effective GenAI tools based on adaptability, accuracy, usability, and integration, to enhance attention, engagement, and academic performance for adults with ADHD.

3.2.1 Explored AI models and tools for UI generation. The research team examined seven common AI models to generate user interfaces, including DALL-E, Flux AI, Vercel VO, Claude AI, and ChatGPT 4.0, using three custom-designed prompts (Figure 10). A survey of 25 UI/UX specialists and non-experts evaluated each model using the criteria accuracy, detail, artistic, relevance and creativity. Vercel VO had the highest score, but ChatGPT and Claude AI also performed well. As a result, the study used a mixed strategy, including Vercel VO for creativity, Claude for long-term reasoning, and ChatGPT for structured responses, to ensure balanced performance while reducing the risk of model-specific errors and blind spots.

3.3 Design and Development

This phase of the research included designing and developing three LMS prototypes: a conventional LMS, a manually optimised ADHD-friendly LMS, and an AI-driven LMS built with enhanced prompting. The design process used Nielsen's Usability Heuristics, HCI theories, UI/UX best practices, and ADHD-specific design concepts in an iterative manner, integrating expert and user feedback. The subject "Philosophy of Science" was chosen as the primary content to reduce bias from existing knowledge, ensuring that user interaction, engagement, and quiz scores accurately represented each prototype's design and educational effects.

3.3.1 Design guidelines and features of ADHD friendly design. The research utilised ADHD-friendly design principles based on the comprehensive literature review, with a particular focus on improving usability for neurodiverse users. Key guidelines were derived from McKnight's 15 design principles[27], which are consistent with general usability standards, and further strengthened by Neuro-Inclusive UX Design Principles[47] and Phalke and Shrivastava's[40] visual design classification for Health Informatics apps. These design elements were used throughout the development of the learning interfaces to ensure that they effectively addressed the unique cognitive and behavioural needs of ADHD users.

3.3.2 Usability Heuristics and HCI principles. The prototypes were designed and implemented using existing HCI standards and UI/UX best practices to ensure usability and accessibility for students with ADHD. Jakob Nielsen's 10 usability heuristics[29, 32] were among the key principles, along with guidelines for error messages[33], accessibility[18], visual design, and user-friendly colour schemes. These principles were integrated and refined based on literature review findings and feedback from UI/UX specialists, ensuring that the interfaces were both effective and inclusive of neurodiverse learners.

3.3.3 Structured Prompting Approach. Guided by an NIH Stage Model + co-design framework for ADHD-oriented digital interventions [41], we specify a reproducible prompt-engineering pipeline for AI-generated LMS components used in our study (4).

Prompting pipeline (inputs → outputs).

- (1) *Stage 0 – Exploratory zero-shot* (behavioural goals, user needs) → conceptual page/flow skeletons.
- (2) *Stage 1A – Constraint prompts* (functional and visual requirements) → feasible, testable UI variants.
- (3) *Stage 1B – Accessibility & persona prompts* ("ADHD-friendly", contrast/focus, progressive disclosure; persona = ADHD students) → neurodiversity-aware artefacts.
- (4) *Stage 3 – Deployment templates* (frozen prompt schema + logging of model/temperature/top-p) → fidelity across classroom runs and analyses.

Template (standardised). Design an [ADHD-friendly] [Component] with [Accessibility Feature(s)] and [Interaction Flow]; optimise for [Behavioural Objective]; return [Output Format].

3.3.4 Prototype Development – Prototype 01: Base LMS. The first prototype developed was a Base LMS (Figure 12), which was used as a control model for an ordinary learning management system used by the general student population. It was built with React

and Tailwind CSS and contained basic features such as login, course listings, a calendar, and assignment tracking, but there were no ADHD-specific design elements or accessibility modifications. Its purpose was to provide a standard for assessing adequately traditional LMS platforms meet the needs of neurodivergent users. While completely functional, the Base LMS did not address major ADHD factors such as cognitive load reduction or user feedback.

3.3.5 Prototype Development – Prototype 02: Manually Improved LMS. The second prototype, the Manually Improved LMS (Figure 2), was developed mainly to assist students with ADHD by including ADHD-friendly design concepts based on research, surveys, and expert remarks. It was built with React and Tailwind CSS and included improvements such as simpler navigation, decreased visual clutter, micro-interactions, focus mode tools, and enhanced typography to minimise cognitive load and increase engagement. Based on Nielsen's Usability Heuristics, HCI principles, WCAG 2.1 standards[51] and McKnight's standards[27], the prototype was enhanced through iterative input from ADHD users. It acted as a bridge between typical LMS platforms and personalised learning tools, providing a more inclusive and user-friendly experience for neurodivergent students.

3.3.6 Prototype Development – Prototype 03: AI-Generated LMS. The third prototype used the LLMs, - ChatGPT-4, Claude AI, and Vercel VO to automatically develop ADHD-friendly LMS interfaces based on usability heuristics, HCI principles, and WCAG 2.1 requirements (Figure 3). A customised ChatGPT-4 plugin was created to generate low-distraction, neurodiversity-aware UI components with constraint-based prompts. Key features were high-contrast navigation, customisable themes, progressive disclosures, and accessibility controls. Despite initial challenges such as a lack of understanding of system architecture, excessive colour use, and uneven API mapping, feedback loops considerably improved results. The experiment showed LLMs' potential and limitations in developing inclusive, AI-powered educational systems for neurodivergent users.

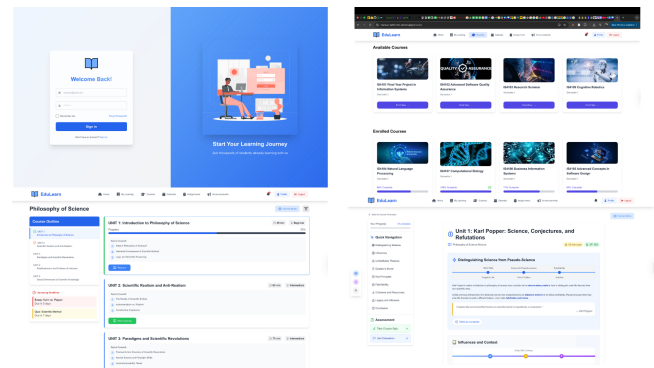


Figure 2: Manually Improved LMS

3.4 Demonstration

The research followed an iterative demonstration phase involving 22 ADHD participants to evaluate three LMS prototypes—Base,

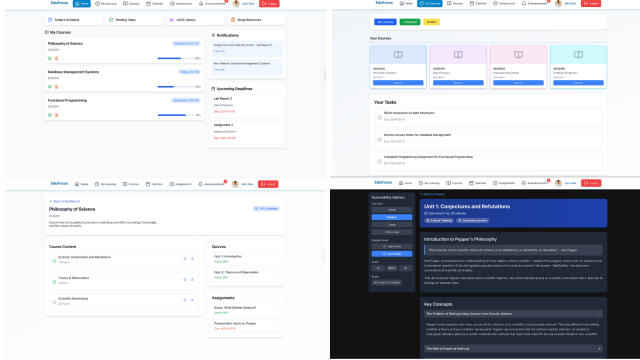


Figure 3: AI-Generated LMS

Manual, and AI-driven. All participants began with the Base LMS to establish a baseline. To avoid evaluation bias, the order of the remaining prototypes was counterbalanced. After using each system, participants completed standardized usability, user experience, and performance (quiz score) assessments, along with follow-up interviews. This mixed-method approach enabled a thorough, real-world evaluation of each system's usability, accessibility, and educational effectiveness for students with ADHD.

3.5 Evaluation

Using data from the demonstration phase, the researchers evaluated the usability and effectiveness of the three LMS prototypes. The data gathered during the demonstration phase were analysed to determine changes in engagement, effectiveness, usability, and learning outcomes. Ethical standards were strictly followed, with informed consent obtained and participant data anonymised. The study ensured the safety, rights, and privacy of all ADHD student participants during the research.

4 Evaluation and Results

4.1 Participants

Twenty-two students with ADHD (ages 18–30), all with prior experience using online learning platforms, participated. To reduce bias, recruitment spanned varied educational backgrounds. The cohort comprised 10 female (4 medically diagnosed; 6 self-diagnosed via the Mental Health America ADHD screening tool [2]) and 12 male participants (4 medically diagnosed; 8 self-diagnosed via the same tool). ADHD severity was assessed using the Adult ADHD Self-Report Scale (ASRS v1.1), developed in collaboration with the World Health Organization; participants in the high positive range were excluded to improve reliability. Figure 13 shows the severity-score distribution, and Table 5 summarises demographics. The near-balanced gender distribution supports objective comparison across the three LMS prototypes (Refer Table 5).

4.2 Usability of the System

Usability of each prototype was measured mainly using the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ) with 22 ADHD users. The SUS questionnaire was shared with our user group, and based on their scores, statistical analysis

methods were applied. Random attention check questions were included to avoid careless answers and ensure accurate ratings. In this study, participants scored on prototypes using UEQ, highlighting variations in satisfaction across them, through 6 scales which are Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, Novelty. The results were used to compare usability through comparative statistical analysis. Random attention check questions were included to ensure reliability and accuracy in the responses.

4.3 Effectiveness of the System

The study compared the effectiveness of three developed systems based on learning outcomes and user engagement, aiming to evaluate how each system helped students with ADHD maintain attention and achieve academic goals during the learning process.

4.3.1 Learning Outcomes - Quiz Performance Analysis. To assess learning outcomes, participants completed standardized 10-question quizzes (multiple-choice, fill-in-the-blank, and true/false) designed to evaluate their understanding and retention of the content.

4.3.2 User Engagement—Task Analysis via Microsoft Clarity. User engagement was assessed through detailed task analysis using Microsoft Clarity, which is a behavioral analytics tool that captures real-time user interaction data and provides insights into how users interact with web-based platforms. Through features such as session recordings, heatmaps, click tracking, and scroll depth analysis, we were able to observe how users navigated each system.

4.4 Experiment 1 - Evaluation of Prototype 1 : Base LMS

4.4.1 SUS Score Analysis. The analysis of the Base LMS usability reveals that its median SUS score falls below the acceptable threshold of 68, indicating a clear need for improvement. The wide interquartile range reflected high variability in user responses, suggesting inconsistent user experiences. With a median score of 56.2 and a mean of 58, the overall usability is rated significantly poor. Notably, 40.9% of participants rated the system below 51, classifying it as having poor usability, while only 22.7% rated it as excellent. These findings collectively highlight that the Base LMS fails to meet usability standards for most users.

4.4.2 UEQ Results Analysis. This reveals that its median score falls below the acceptable threshold of 4.0, with low overall scores and high variance, indicating inconsistent user experiences. The presence of outliers with particularly low scores further contributes to the variability. Among the six UEQ dimensions, the lowest mean scores were observed in Stimulation (3.60) and Novelty (3.58), suggesting the system lacks engagement and innovation. Only Dependability (4.19) and Perspicuity (4.07) achieved scores above the threshold, indicating that the remaining dimensions require significant improvement. Additionally, quiz score analysis showed high variability, implying inconsistency in learning outcomes and difficulties in knowledge retention and application among users.

4.5 Experiment 2 – Evaluation of Prototype 2: Manually Improved LMS

This experiment evaluated an LMS, modified by designers without using artificial intelligence, tailored for ADHD users, incorporating design adjustments from Experiment 1 to reduce cognitive load and enhance focus.

4.5.1 SUS Scores Analysis. The improved LMS achieved a median SUS score of 75 and a mean of 73.8, both above the usability threshold of 68. The distribution was left-skewed, indicating that most participants rated the system positively. Notably, 36.4% rated its usability as excellent (score > 80.3), reflecting significant improvements over the base LMS.

4.5.2 UEQ Results Analysis. UEQ scores remained mixed. The median was below the acceptable threshold of 4.0, with high variance across responses. Only *Dependability* (4.07) and *Perspicuity* (4.09) surpassed the 4.0 mark. *Novelty* received the lowest mean score (3.76), indicating limited perceived innovation. Further design iterations are needed to enhance overall user experience.

4.5.3 Learning Outcomes. When measuring the quiz scores acquired for Manual LMS the results indicate an improved median quiz score and reduced variance compared to Base LMS. Further, the figure 11, indicates the mean is 6.5, which is comparatively higher than the value acquired in Base LMS, implying improved comprehension and retention.

4.6 Experiment 3 – Evaluation of Prototype 3: AI-Generated LMS

This experiment evaluated the AI-generated LMS prototype, designed using AI-driven UI generation techniques such as HCI-informed prompting and ADHD-specific feedback.

4.6.1 SUS Scores Analysis. The AI LMS achieved the highest usability among all prototypes, with a median SUS score of 83.8 and a mean of 80.1—well above the acceptable threshold of 68. Notably, 63.6% of participants rated it as having excellent usability (score > 80.3), indicating strong user satisfaction and preference over previous prototypes.

4.6.2 UEQ Results Analysis. Despite the high usability, the median UEQ score remained slightly below the threshold of 4.0. However, the AI LMS recorded higher scores across all six dimensions compared to previous versions. Notable improvements were seen in *Dependability* (4.31) and *Novelty* (4.07), both exceeding the threshold, while *Attractiveness* had the lowest score (3.80), suggesting room for enhancement in visual appeal and engagement.

4.6.3 Learning Outcomes. Participants using the AI LMS demonstrated improved learning performance, with a median quiz score of 6.0 and a mean of 5.9. This reflects better consistency and higher retention than with the Base LMS, and performance comparable to the Manual LMS. The reduced score variance suggests more uniform learning effectiveness among users.

4.7 Overall System Evaluation

4.7.1 Descriptive statistics Analysis. Descriptive statistics provide a foundational understanding of central tendencies (mean) and

dispersion standard deviation (SD) of the data. When considering SUS scores, Base LMS (Mean=57.95) falls below the acceptable threshold (68) while Manual LMS (Mean=73.75) and AI LMS (Mean=80.11) are higher than the acceptable threshold. Manual LMS shows a significant improvement compared to Base LMS, while AI LMS (Mean=80.11) achieves the highest. Base LMS shows highest SD (23.35), indicating inconsistent user experiences while AI LMS has the lowest SD (17.16), demonstrating reliable performance. By analysing the range of SUS scores, Base LMS exhibits extreme scores (20-97.5) while AI LMS maintains tighter range (45-100). According to these statistics of SUS scores, Base LMS with the lowest mean, indicates poor usability, Manual LMS shows a moderate increase in usability and AI LMS with the highest mean SUS, indicates exceptional usability and user satisfaction.

All systems show mean UEQ scores near neutral (4.0) but, Base LMS (3.89) and Manual LMS (3.91) fails to meet positive UX threshold. AI LMS (4.00) is the first to cross neutral benchmark. When considering the inter quartile ranges, Base LMS has widest UX perception variability while AI LMS shows most consistent positive ratings with less variance compared to other two prototypes. Therefore the AI LMS scored consistently high across user experience dimensions. Further the Manual LMS improved upon the Base LMS but did not reach AI LMS levels.

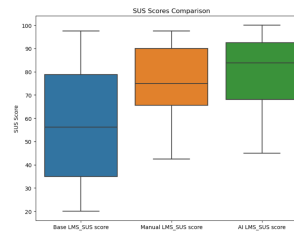


Figure 4: SUS Scores Comparison

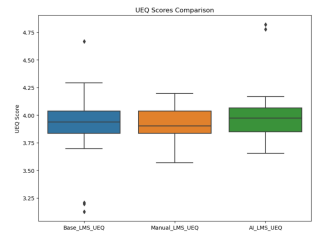


Figure 5: UEQ Scores Comparison

4.7.2 Comparison Visualizations Analysis. Figure 4 states AI LMS achieved a median SUS score above 85, the Manual LMS around 72, and the Base LMS below 60. Therefore this comparison indicates that AI LMS has achieved “Excellent” SUS score, while Base LMS falls below 60 in the “Poor” category, implying that AI LMS achieved the highest usability.

According to Figure 5, when comparing the UEQ scores, similar trends can be observed, AI LMS excelling in all dimensions. However, here it is not much variant across three LMS as in SUS score distribution. All the prototypes indicates median scores near the threshold while AI LMS achieves the highest score ensuring more usability.

These Measures showed that the AI LMS has the highest mean and least SD in SUS scores, indicating consistent user ratings UEQ score mean values of three prototypes show no significant variability, although AI LMS mean score is somewhat higher. Manual LMS SD seems to have the lowest UEQ score.

4.7.3 Normality Test. We performed Shapiro-Wilk Test (Normality Testing) [25] to check if the SUS & UEQ values of Base LMS, Manual LMS, AI-generated LMS are normal or non-normal.

Figure 7 shows the Shapiro-Wilk Test Results we derived. P-values for most of the data sets are less than 0.05, indicating that the data significantly deviate from a normal distribution. But UEQ of Base LMS and AI LMS derived as non-normal.

According to the below normality test results, although most of the SUS & UEQ results of the LMS are normal, there are some non-normal values as well. Further in the evaluations of this study, we are using repeated values by using same user group for rating all the three LMSs. Therefore we cannot use One way ANOVA, and as an alternative for non-parametric testing, we used Friedman Test for further analysis.

Friedman Tests:

Friedman Test (SUS): $\chi^2=3.805$, $p=0.1492$

Friedman Test (UEQ): $\chi^2=1.238$, $p=0.5385$

Figure 6: Friedman Test Results

Normality Test Summary:

Measure	LMS	W	p-value	Normality
0	SUS Base	0.938631	0.185452	Normal
1	UEQ Base	0.903014	0.034168	Non-normal
2	SUS Manual	0.916413	0.064091	Normal
3	UEQ Manual	0.953035	0.362282	Normal
4	SUS AI	0.914943	0.059774	Normal
5	UEQ AI	0.778409	0.000234	Non-normal

Figure 7: Normality Tests (Shapiro-Wilk)

4.7.4 Friedman Test. Friedman Test [11], a non-parametric test, was used to assess statistical differences in SUS and UEQ scores across the three LMS prototypes.

- **Null Hypothesis:** No significant differences in SUS or UEQ scores across LMS variants.
- **Alternative Hypothesis:** At least one LMS variant differs significantly.

Figure 6 shows the p-values: SUS ($p = 0.149$) and UEQ ($p = 0.538$). Both values exceed the 0.05 threshold, indicating no statistically significant differences in usability or user experience across the LMSs.

Despite non-significant results, post hoc Wilcoxon signed-rank tests [22] were conducted for pairwise comparisons due to the small sample size and potential subtle differences.

4.7.5 Post-hoc Analysis. To identify which specific pairs of LMS differ significantly, pairwise comparisons were made using the Wilcoxon signed rank test [22]. Bonferroni correction was applied to control for Type I error due to multiple comparisons.

Table 2: Post-hoc Analysis (Wilcoxon Signed-Rank Tests with Bonferroni Correction)

SUS Scores				
Comparison	W	p	d	adj_p
Base vs Manual	69.5	0.0634	-0.576	0.190
Base vs AI	38.5	0.0029	-0.808	0.009
Manual vs AI	33.5	0.0734	-0.403	0.220
UEQ Scores				
Comparison	W	p	d	adj_p
Base vs Manual	120	0.849	-0.051	1
Base vs AI	72.5	0.225	-0.274	0.675
Manual vs AI	98.5	0.808	-0.245	1

Table 3: Correlation Analysis—Spearman's ρ between SUS and UEQ

Condition	Spearman's ρ	p-value
Base	0.373	0.0870
Manual	0.263	0.2365
AI	-0.158	0.4836

Table 2 displays the results of each pairs of LMS with adjusted p values. It shows that SUS score of, Base LMS vs. AI-Generated LMS pair is $p = 0.0085$, representing a statistical significant difference ($p < 0.0167$). All the other pairs are not performing statistically significant difference ($p < 0.0167$). It ensures that Base vs AI comparison reached statistical significance with large effect size ($d = -0.81$) confirming the practical importance.

Post-hoc comparisons of the SUS scores revealed a statistically significant difference between the Base LMS and the AI-enhanced LMS, with a large effect size, indicating superior usability in the AI-driven system. In contrast, post-hoc comparisons of the UEQ scores did not yield any statistically significant differences between system pairs. These results imply that AI LMS shows statistically better usability than conventional LMS. Manual optimizations alone may not be sufficient for significant ADHD usability improvements. As well as it shows that while AI improves usability with SUS, core UX dimensions with UEQ still remain similar across systems.

4.7.6 Correlation Analysis. Spearman's rank correlation analysis [12] was used to examine the relationship between System Usability Scale (SUS) and User Experience Questionnaire (UEQ) scores across three LMS.

According to table 3, Base LMS shows $\rho = 0.373$ ($p = 0.087$) indicating a moderate positive relationship. It fails to reach significance ($p < 0.005$) may due to high variability in ADHD responses and limited sample size ($n=22$). Manually improved LMS shows $\rho = 0.263$ ($p = 0.237$), indicating a weak positive relationship (not significant). AI LMS represents $\rho = -0.158$ ($p = 0.484$), indicating a negligible inverse relationship (not significant). According to the above figures, there were no significant correlations found between SUS and UEQ scores.

4.7.7 Analysis of LMS Evaluation Scores Across ADHD Levels. To perform a better analysis of how the ADHD levels of the users have affected for the evaluations of each prototype, we compared the three LMSs — Base, Manual, and AI LMS based on user feedback from participants across different ADHD levels using usability and effectiveness metrics.

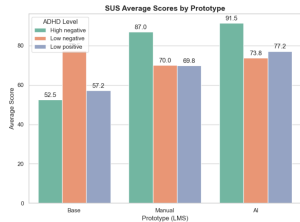


Figure 8: Analysis of SUS Scores Across ADHD Levels

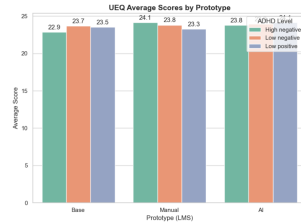


Figure 9: Analysis of UEQ Scores Across ADHD Levels

As shown in Figure 8, average SUS scores varied across ADHD levels and LMS types. Participants with **low negative ADHD traits** gave the **highest SUS scores to the Base LMS**, indicating its suitability for the general population. Manual and AI LMS were rated higher by **high negative individuals**, suggesting improved usability for users with stronger ADHD symptoms.

Similarly, Figure 9 shows that while there were **no significant differences in UEQ scores**, the trends mirrored SUS results. The Base LMS was again preferred by low negative individuals, while **Manual and AI LMS received higher UEQ scores from high negative individuals**, indicating better alignment with ADHD user needs.

Figure 11 shows the comparison of the average Quiz scores of different ADHD levels variation across each of the LMS. Overall it represents most of the low negative individuals have acquired highest scores across all of the three LMS. Low positive individuals have scored comparatively more than high negative individuals across all the LMS. It implies that quiz performance has not been much positively supported for ADHD population.

4.7.8 Microsoft Clarity Analysis. User interaction metrics collected via Microsoft Clarity were analyzed across three LMS prototypes to evaluate usability and engagement, with particular focus on ADHD users.

Prototype 1 (Base LMS) exhibited limited sustained engagement. Attention was concentrated at the top of content pages, suggesting cognitive fatigue. Users viewed an average of 6.7 pages per session and reached 90.4% scroll depth. However, only 28% of the session time was spent actively, with 26.85% dead clicks and 36.11% quick backs, indicating user frustration and navigation issues. The performance score was 77/100, hampered by the slowest loading time (LCP 7.6s) and highest interaction delay (INP 250ms).

Prototype 2 demonstrated significant improvements. Attention was evenly distributed across pages, indicating improved focus retention. Users visited an average of 8.62 pages per session, with an active time of 40% (approximately 4.5 minutes). Notably, there were 0% rage clicks, only 11.76% dead clicks, and 23.53% quick backs. Performance improved with a score of 82/100, featuring the fastest

loading time (LCP 4.5s), flawless layout stability (CLS 0), and a reduced INP of 190ms.

Prototype 3 showed high exploratory activity, with 8.52 pages per session and the highest scroll depth (94.54%). It achieved smooth interaction (INP 150ms) and excellent layout stability (CLS 0.001). However, active engagement dropped to 21.6%, and interaction friction increased with 4% rage clicks, 32% dead clicks, and 56% quick backs. The performance score was 80/100, with a slightly improved LCP of 6.5s. These issues were likely due to overwhelming or inconsistent AI-generated content.

Comparative results highlight Prototype 2 as the most effective LMS version. It delivered the best balance of usability, engagement, and system performance, effectively addressing ADHD-specific needs through clear layouts and interactive design elements. While Prototype 3 provided personalized content and high engagement potential, its complexity reduced usability for neurodivergent learners, emphasizing the need for thoughtful AI integration.

4.7.9 Thematic Analysis of Participant Feedback. A brief inductive synthesis of all participant comments surfaced four recurrent themes: **Orientation & Feedback** — Requests for clearer navigation and results transparency (“No back button in the system”; “Please show which were the right and wrong answers in quiz”). **Focus vs. Distraction** — Split reactions to animations, focus mode, and the study timer (“amazing”; “good addition” vs. “too distracting”). **Progressive Disclosure** — Positive reception to staged content (“The way the content was presented was good, not showing everything at once”), supporting reduced overload. **Content Fidelity & Oversight** — AI-generated images/interactive elements sometimes lacked contextual relevance and required human correction. These themes informed the iterative improvements in both Manual and AI prototypes and contextualise why engagement patterns diverged from usability scores.

4.8 Recommendations

The AI-Generated LMS. was a significant challenge to develop during this research due to the limited capacity of LLMs to comprehend the system architecture of a web application. This constraint led to architectural inconsistencies, error propagation, and a reduction in system reliability. The AI-Generated LMS. necessitated substantial human supervision and architectural guidance during its development. Future research directions that concentrate on reference architectures, iterative approaches, and architectural guidance systems have the potential to overcome these constraints. It is advisable to employ an AI model that is aware of the web system’s architecture in future research and test with a larger, more diverse ADHD cohort to improve interpretive clarity and generalizability.

4.9 Future Work

Building on these findings, several promising avenues can enhance personalized learning interfaces for ADHD students. Integrating eye tracking and clickstream analysis can help identify design elements that improve attention and task completion. Longitudinal studies could assess the long-term impact of Manual and AI-driven LMSs on knowledge retention and academic performance. Wearable biometric sensors (e.g., EEG headsets, heart-rate monitors, and galvanic skin response sensors) could provide real-time feedback

to dynamically adjust attention-retention strategies. Interfaces may also be personalized based on real-time behavioral analysis to align with individual neurocognitive profiles, enabling adaptive, multi-modal learning experiences.

Future research could develop specialized AI models tailored to ADHD learners, offering more effective support than general-purpose models. Larger studies with medically diagnosed ADHD students are essential to better understand their cognitive challenges and refine personalized educational interventions.

4.10 Discussion

This study examined the impact of GenAI-driven customization versus manual configuration on LMS usability and effectiveness for students with ADHD. Three LMS prototypes were developed and evaluated on effectiveness, usability, engagement, and learning outcomes. The GenAI-driven LMS demonstrated superior usability, supported by quantitative results and comments such as “This is much more user-friendly” and “Much better UI.” The Manual LMS slightly outperformed the AI-driven LMS on effectiveness, with participants praising animations, focus mode, and the study timer (e.g., “The graphics and animations were amazing,” “The study timer was a good addition”). Notably, some users found these features distracting, indicating a need to balance stimulation and cognitive load in future designs. Addressing the first two secondary research questions, the baseline LMS exhibited clear usability limitations in navigation and feedback. These insights guided improvements in the Manual and AI-driven prototypes, demonstrating the value of iterative, user-informed design. The thematic analysis (Section 4.7.9) helped us explain why the Manual LMS achieved stronger engagement despite the AI-LMS’s superior usability.

4.10.1 Why High Usability Did Not Maximise Engagement? Despite higher perceived usability, the AI-generated LMS elicited weaker engagement than the Manual LMS. Three mechanisms apparent in participant feedback help explain this divergence: (i) *Hedonic stimulation vs. clarity*—the Manual LMS’s animations, focus mode, and timer provided motivating cues, whereas the AI UI emphasised low-distraction clarity that aids learnability but offers fewer stimulating rewards; (ii) *Attention scaffolding*—timers/focus mode and progressive pacing increased momentum for some but were “too distracting” for others, revealing a trade-off between arousal and cognitive load; and (iii) *Content fidelity*—AI-generated images/interactive elements occasionally lacked contextual relevance and required human oversight, tempering enthusiasm despite ease of use.

The third secondary question focused on optimizing generative AI prompting strategies to improve engagement and accessibility. Prompt engineering enabled the development of user-centered, contextually relevant interfaces. Features such as progressive content disclosure and interactive quiz reviews were well received, with comments including ‘The way the content was presented was good, not showing everything at once’. However, limitations in AI-generated content, particularly images and interactive elements, presented challenges, which required significant human oversight due to inaccuracies and lack of contextual relevance.

Finally, while AI-driven LMS achieved higher usability, manual LMS showed slightly better learning outcomes, as reflected in quiz

scores. These findings underscore the potential of GenAI to improve personalization and usability in educational technologies. However, realizing its full benefits will depend on more advanced prompting methods and improved content generation capabilities. Overall, the study affirms the promise of GenAI in supporting neurodiverse learners, while also highlighting the importance of thoughtful human-centered design and evaluation.

4.11 Conclusion

This research aimed to explore how Generative AI can enhance personalized learning interfaces for students with ADHD (ages 18–30), improving their academic experience and promoting equity in education. Through literature reviews, surveys, and user feedback, we identified the unique learning challenges ADHD students face. We evaluated three LMS prototypes: a standard Base LMS, a manually optimized LMS tailored for ADHD users, and an AI-generated LMS adapted using enhanced prompting techniques. A user group of 22 ADHD individuals participated in this evaluation. To assess usability and effectiveness, we collected SUS and UEQ responses and analyzed quiz scores and Microsoft Clarity results. Statistical methods, including normality tests, Friedman tests, Wilcoxon Signed-Rank tests, and Spearman’s correlation, were applied for deeper insights.

Results showed that the AI LMS achieved the highest usability scores across SUS and UEQ metrics, particularly excelling in novelty and engagement. The Manual LMS had moderate usability and outperformed both other systems in effectiveness, based on quiz scores and clarity analytics. The Base LMS showed the lowest performance overall and was only preferred by users with fewer ADHD symptoms. In conclusion, the AI LMS proved most usable, while the Manual LMS was most effective in supporting learning outcomes. Both tailored systems significantly outperformed the Base LMS for ADHD users, underscoring the importance of personalized learning environments in addressing their specific educational needs.

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Appendix

- Primary GitHub Repository with all the resources used : <https://github.com/RashmiNirasha/GenAIRsearch.git>

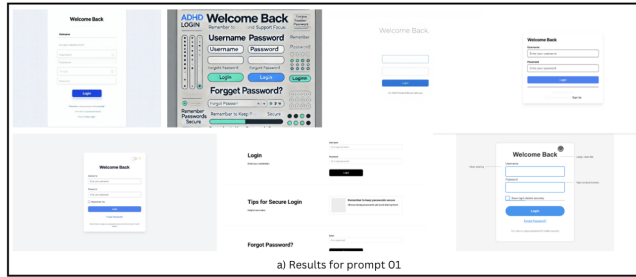


Figure 10: Wireframe Generated for Test 01

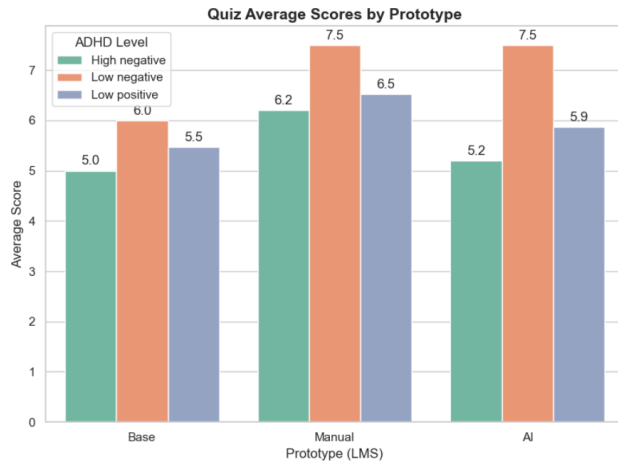


Figure 11: Analysis of Quiz Scores Across ADHD Levels

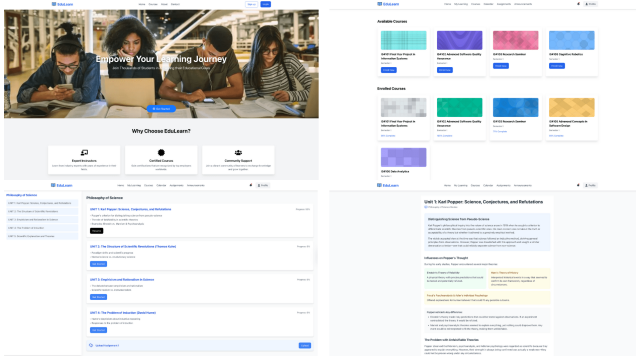


Figure 12: Base LMS

Table 4: Representative Prompt Evolution

Stage	Example
Initial (zero-shot)	Generate a course page for the learning management system.
Refined (constraints)	Create a profile page with editable fields, full-width layout, an editable picture, and a save action.
Final (accessibility + persona)	Design an ADHD-friendly profile page with explicit focus states, accessible colors, editable fields/picture, and a way to save feedback for ADHD students.

Table 5: Demographic profile of participants (N = 22)

Characteristic	Category	Count (n)	Percent (%)
Age range	18–30 years	22	100
Gender	Female	10	45.5
	Male	12	54.5
ADHD diagnosis type	Medically diagnosed	7	31.8
	Self-diagnosed (MHA tool)	15	68.2
ASRS v1.1 severity	Low negative	3	13.6
	High negative	4	18.2
	Low positive	15	68.2
	High positive (excluded)	0	0.0
Education level	High school	1	4.5
	Undergraduate	20	90.9
	Postgraduate (Master's)	1	4.5
Prior LMS experience	All participants had prior experience	–	–

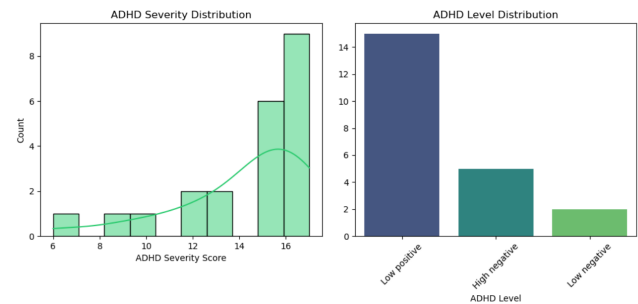


Figure 13: ADHD Severity Level Distribution